

Level 20 → Level 21

nc -l & 명령어로 백그라운드에서 포트를 열어놓고 현재 비밀번호를 전송시킨다. ./suconnect로 포트에 연결하니 다음 비밀번호를 얻을 수 있었다.

```
bandit20@bandit:~$ ls
suconnect
bandit20@bandit:~$ ./suconnect
Usage: ./suconnect <portnumber>
This program will connect to the given port on localhost using TCP. If it receives the correct password from the other side, the next password is transmitted back.
bandit20@bandit:~$ echo 0qXahG8Zj0VMN9Ghs7iOWsCfZyX0UbY0 | nc -l -p 12345 &
[1] 25
bandit20@bandit:~$ ./suconnect 12345
Read: 0qXahG8Zj0VMN9Ghs7iOWsCfZyX0UbY0
Password matches, sending next password
Ee0ULMCra2q0dSkYj561DX7s1CpBu0Bt
```

Level 21 → Level 22

문제에서 제시한대로 /etc/cron.d/를 확인하니 cronjob_bandit22라는 파일이 보인다. cat 명령어로 열어보니 비밀번호를 갖고 있는 경로가 확인된다.

```
bandit21@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup  otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job  sysstat
bandit21@bandit:~$ cat /etc/cron.d/cronjob_bandit22
@reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
* * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
bandit21@bandit:~$ cat /usr/bin/cronjob_bandit22.sh
#!/bin/bash
chmod 644 /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
cat /etc/bandit_pass/bandit22 > /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
bandit21@bandit:~$ cat /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q
```

Level 22 → Level 23

21과 비슷하게 스크립트를 확인한다.

```
bandit22@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup  otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job  sysstat
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"

cat /etc/bandit_pass/$myname > /tmp/$mytarget
```

스크립트에 시키는대로 입력하니 해시값을 알려준다. 그 해시값을 경로로 삼아 비밀번호를 얻을 수 있었다.

```
bandit22@bandit:~$ echo I am user bandit23 | md5sum | cut -d ' ' -f 1
8ca319486bfbbc3663ea0fbe81326349
bandit22@bandit:~$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349
0Zf11ioIjMVN551jX3CmStKLYqjk54Ga
```

Level 23 $\rightarrow$ Level 24

21, 22와 마찬가지로 스크립트를 확인할 수 있다.

```
bandit23@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup  otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job  sysstat
bandit23@bandit:~$ cat /etc/cron.d/cronjob_bandit24
@reboot bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
* * * * * bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
bandit23@bandit:~$ cat /usr/bin/cronjob_bandit24.sh
#!/bin/bash

shopt -s nullglob

myname=$(whoami)

cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo:"
for i in * .*;
do
    if [ "$i" != "." ] && [ "$i" != ".." ];
    then
        echo "Handling $i"
        owner="$(stat --format "%U" "$i")"
        if [ "${owner}" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 9 60 "$i"
        fi
        rm -rf "$i"
    fi
done
```

임시 디렉터리를 만들고 권한을 부여한다. 위에서 읽은대로 스크립트를 만들고 여기도 권한을 부여한다. 원래 디렉터리에 복사하고 기다린 후 password 파일을 읽으니 비밀번호가 나왔다.

```
donebandit23@bandit:~$ mkdir /tmp/asdf
bandit23@bandit:~$ chmod 777 /tmp/asdf
bandit23@bandit:~$ nano /tmp/asdf/a.sh
Unable to create directory /home/bandit23/.local/share/nano/: No such file or directory
It is required for saving/loading search history or cursor positions.

bandit23@bandit:~$ chmod 777 /tmp/asdf/a.sh
bandit23@bandit:~$ cp /tmp/asdf/a.sh /var/spool/bandit24/foo/
bandit23@bandit:~$ cat /tmp/asdf/password.txt
gb8KRRcSshuZXI0tUuR6yp0FjiZbf3G8
```

Level 24 $\rightarrow$ Level 25

30002 포트에 4자리 숫자로 브루트포스를 시도하라고 한다. 임시 디렉터리를 만들어 브루트포스를 위한 스크립트를 만들고 30002 포트에 전달한다.

[illegible]

쪽 내려보니 비밀번호를 확인할 수 있다.

```
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Wrong! Please enter the correct current password and pincode. Try again.  
Correct!  
The password of user bandit25 is iCi86ttT4KSNe1armKiwbQNmB3YJP3q4
```